Project proposal

DSA 101 Introduction to Data Science · Fall 2026 · Stage 1 of 3

Presented	Monday 19 October, in the lecture slot: **5 minutes to present, 5 minutes of questions**
Hand in	The one-page template (page 2) as a PDF, by **midnight before: Sunday 18 October, 23:59**
Weight	3% of the course grade (project in total: 18%)
Team	Your team of three. One proposal per team. Everyone presents a part.

What the proposal is for

This project is meant to be hard. You may use AI tools, so a question that one prompt can answer is too easy. The proposal shows that your team has **a dataset you can load**, **a question the data could say no to**, and **a realistic plan**. You should be able to explain every part of it yourselves. Ambition comes later: see “What we expect later” below.

What you must show

- **Dataset, loaded.** Name, source, and the rows and columns you got when you loaded it. What does one row represent?
- **Question.** One question, written so the data could answer no. “What affects house prices?” is a topic; “Do coastal listings cost more than inland ones after accounting for size?” is a question.
- **First look.** One figure or number from your own data, with what it shows and what it does not show.
- **Beyond the course (recommended, not required now).** If you already have an idea for a method we have not taught, write it down. If not, write “not yet”. It is not marked at this stage.
- **Biggest risk.** Missing data, a biased sample, a column that leaks the answer, a licence, a source that may break. What will you do about it?
- **Plan and roles.** What you will have at the checkpoint (23 November), and who owns what.

What we expect later

Since AI tools are allowed, we expect the finished project to go beyond what we teach in class: at least one method or idea that is new to you, used because it fits your question and explained in your own words. **We do not expect this in the proposal.** Dataset and question matter most now. If an idea below already appeals to you, mention it; it helps us advise you early.

Ideas to start from	Use it for
PCA / dimensionality reduction	Many correlated columns; finding the few directions that carry the variation
Explainable AI: permutation importance, SHAP, partial dependence	Opening up a model: which features drive it, and does that make sense
Clustering beyond k-means; anomaly detection	Finding groups or outliers without labels, and checking they are real
Gradient-boosted trees, regularisation paths	Stronger models, with an honest baseline and honest evaluation
Causal thinking: confounders, matching, difference-in-differences	Asking whether X *changes* Y, not only whether they move together
Time series: decomposition, forecasting with proper splits	Trend and seasonality; not leaking the future
Your own idea	Agree it with the instructor in advance

How the five minutes might go

Slide	Content	About
1	Question, and why it is interesting	1 min
2	Dataset: source, size, what a row is	1 min
3	First look: your figure or number, read honestly	1 min
4	Biggest risk, and an idea to go further if you have one	1.5 min
5	Plan and roles	0.5 min

Use at most 5 slides. In the five minutes of questions, **anyone on the team can be asked about any part**, including the data, the question and any method you mention.

Rules

- Use any dataset you may legally use, and cite it. Avoid Iris, Titanic, Airbnb NYC and MNIST: their answers are everywhere.
- The question may change after the proposal. Tell us at the checkpoint what changed and why.
- AI tools are allowed under the course AI-use policy. Keep your AI usage log.

Proposal template

One page, hand in as PDF by Sunday 18 October, 23:59. Short answers; replace the hints with your own words.

Team name	
Members (name, student number)	

1 · Question

One sentence. Then one sentence: what result would make the answer no?

2 · Dataset

Name, source and link, licence. Rows × columns as you loaded it. What does one row represent?

3 · First look

Paste one figure or number from your own data. Caption: what it shows, and what it does not show.

4 · Beyond the course (optional)

An idea you may explore later, and why it might fit. Or write “not yet”. Not marked.

5 · Biggest risk

What could go wrong, and what will you do about it?

6 · Plan and roles

What you will have by 23 November, and who owns which part.